

MHD QAIS ALSALEH

Software Developer | Full-Stack | Cloud | Research | JavaScript | React | Python | .NET
qaisalsaleh251@gmail.com | +27 630 116 824 | linkedin.com/in/mhd-qais-alsaleh | github.com/qais-mle7y | Pretoria, South Africa

PROFESSIONAL SUMMARY

Software Developer with a BSc in Computer Science from the University of the Witwatersrand and a completed BSc Honours in Computer Science from the University of Pretoria. Strong full-stack experience across React, TypeScript, Node.js, Python/FastAPI, C#, ASP.NET Core, databases, Microsoft Azure, Docker, and CI/CD. Recent honours work includes Data Mining with distinction, Natural Language Processing with distinction, and research on automated assessment of algorithmic thinking using flowcharts. Bilingual communicator in English and Arabic with experience in peer collaboration, technical leadership, and research-informed software development.

TECHNICAL SKILLS

Programming: JavaScript, TypeScript, Python, C#, Java, C++, SQL, HTML, CSS

Frontend: React, Tailwind CSS, responsive design, Three.js, Chart.js

Backend: Node.js, Express.js, FastAPI, ASP.NET Core, Entity Framework Core, REST APIs

Databases: SQL Server, PostgreSQL, MongoDB, SQLite, query optimization

Cloud & DevOps: Microsoft Azure, Docker, GitHub Actions, CI/CD, Git, Linux

AI & Data: Data mining, natural language processing, machine learning, graph representations

Security & QA: JWT, authentication, authorization, RBAC, Jest, unit and integration testing

PROJECTS AND RESEARCH

Flowchart Analysis Tool | Honours Research Project | 2025

- Built a learning tool for automated assessment of algorithmic thinking using flowchart diagrams.
- Parsed flowcharts into graph representations and ran rule-based checks for completeness, decision branching, structural integrity, unreachable paths, and loop termination.
- Generated simplified linear code with explanatory comments to reduce cognitive load for novice programmers.

Fat Loss Tracker | Full-Stack Developer | 2026

- Developed a mobile-first tracker for daily logs, body measurements, progress photos, trend charts, streaks, reminders, and secure authentication.
- Implemented a FastAPI backend with SQLite persistence and a React TypeScript frontend for a structured personal health workflow.

API Rate Limiter Dashboard | Full-Stack Developer | 2026

- Built a sliding-window rate limiting demo with Node.js, Express, React, Chart.js, and Server-Sent Events.
- Created live monitoring views, tiered API keys, and an API tester panel for single and burst request simulation.

Tarkov Quest Finder | Browser Extension Developer | 2025

- Created a Chrome extension for searching Escape from Tarkov quests and viewing objectives, requirements, rewards, and trader details.
- Integrated the tarkov.dev GraphQL API with local caching through Chrome Storage for faster repeat access.

Personal Portfolio Website | Developer | 2024 - 2026

- Built and maintained a responsive React, TypeScript, Vite, and Tailwind portfolio with project, education, research, and CV sections.

EDUCATION

BSc Honours in Computer Science | University of Pretoria | 2025 - 2026

- COS781 Data Mining: 82, pass with distinction.
- COS760 Natural Language Processing: pass with distinction.
- COS700 Research Report: Automated Assessment of Algorithmic Thinking Using Flowcharts.

BSc Computer Science | University of the Witwatersrand | 2022 - 2024

- Relevant Coursework: Data Structures, Software Design, Computer Networks, Databases, Object-Oriented Programming

CERTIFICATIONS

SQL and Business Analysis Certificate — Signature Academy | May 2025

Alibaba Cloud Certified Associate (Cloud Computing and Security)

ADDITIONAL INFORMATION

Languages: English (Advanced), Arabic (Native)

Soft Skills: Problem-Solving, Team Collaboration, Peer Mentoring, Technical Leadership, Self-Directed Learning, Requirements Analysis, Written and Verbal Communication